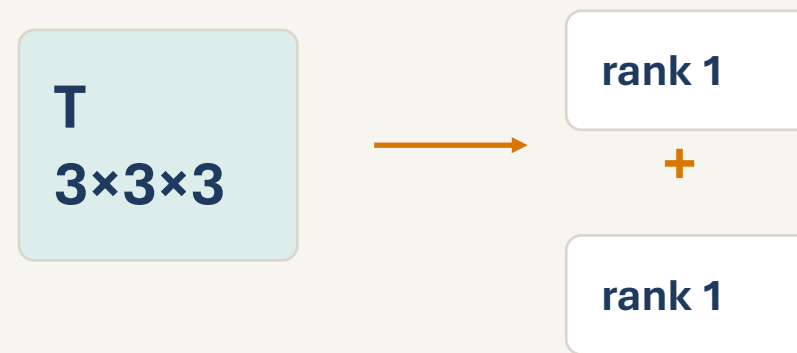


Problem 5

Low-Rank Tensor Approximation

Approximating tensors in $\mathbb{C}^3 \otimes \mathbb{C}^3 \otimes \mathbb{C}^3$ by rank-2 CP tensors

Preliminary Presentation



rank ≤ 2 approximation

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Project goal: from a cube of numbers to two simple patterns

02

Input

$$T \in \mathbb{C}^3 \otimes \mathbb{C}^3 \otimes \mathbb{C}^3$$

A 3×3×3 tensor
(27 complex entries)

Approximation

$$T' = a_1 \otimes b_1 \otimes c_1 + a_2 \otimes b_2 \otimes c_2$$

sum of two rank-one tensors

Goal

minimize
 $\|T - T'\|$

Project interpretation

Design a practical procedure for rank-2 approximation, then implement and evaluate it.

Find $A, B, C \in \mathbb{C}^{3 \times 2}$ such that

$$T'_{ijk} = \sum_{r=1}^2 A_{ir} B_{jr} C_{kr}$$

Optimization problem

$$\text{minimize } \|T - \sum_{r=1}^2 a_r \otimes b_r \otimes c_r\|_F^2$$

Frobenius norm

$$\|T - T'\|_F^2 = \sum_{i,j,k} |T_{ijk} - T'_{ijk}|^2$$

Why this norm?

- measures entrywise reconstruction error
- natural least-squares objective
- simple to compute and compare across trials

Why the problem is nontrivial

04

Matrix case

SVD gives a clean best rank- r approximation.



Tensor case

No simple global SVD-like formula for CP rank approximation.

Main difficulties

**Non-convex
objective**

Local minima

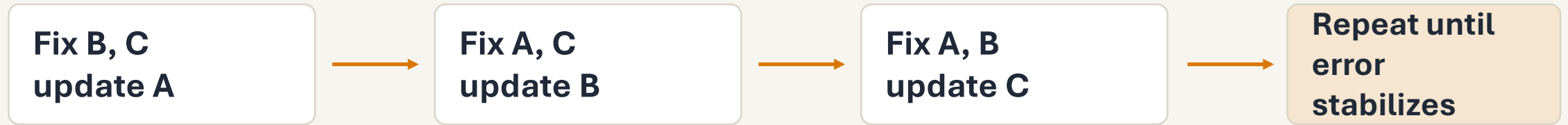
**Initialization
sensitivity**

Project stance We aim for a robust numerical pipeline, not a closed-form theorem.

Proposed method: CP-ALS with multi-start

05

ALS update cycle



Why ALS?

- full problem is nonlinear in A, B, C together
- with two factors fixed, the remaining factor is least-squares
- standard baseline for CP decomposition

Multi-start strategy

Run ALS from many random initializations and choose the smallest relative error.

Planned prototype and evaluation

Implementation plan

- Python + NumPy for tensor generation and error calculation
- TensorLy parafac as the CP-ALS baseline
- optional direct ALS implementation for transparency

Evaluation metric

$$\text{relative error} = \|T - T'\|_F / \|T\|_F$$

Planned test cases

1. Exact rank-2

Sanity check: error should be near zero.

2. Noisy rank-2

Stability test near rank 2.

3. Random tensor

Approximation quality for generic inputs.

Expected output and project roadmap

07

Final output we aim to deliver

- rank-2 CP approximation procedure
- working Python implementation
- experiments on planned test cases
- analysis of convergence and initialization effects

Timeline



Takeaway

We will deliver a practical rank-2 approximation pipeline and evaluate its performance.